Project Name : Fitness tracker data analysis

1. Objectives

- To analyze fitness tracker data and identify patterns in user activity.
- To uncover insights about daily steps, calories burned, and activity intensity.
- To visualize trends and correlations to provide actionable recommendations for improving physical activity and overall health.

2. Problem Statements

- How active are users on a daily basis based on step count and calories burned?
- Are there specific trends in activity levels over time or across weekdays?
- What is the relationship between steps taken and calories burned?
- Can we identify areas for improvement in user activity levels?

3. Data Description

The dataset used for the analysis includes the following columns:

- **ActivityDate**: The date of the activity recorded.
- **TotalSteps**: Total steps taken in a day.
- **Calories**: Total calories burned in a day.
- **VeryActiveMinutes**: Minutes spent in very active exercise.
- **FairlyActiveMinutes**: Minutes spent in moderately active exercise.
- **LightlyActiveMinutes**: Minutes spent in light activities.

Additional columns were derived during the analysis:

- **DayOfWeek**: Extracted from ActivityDate to analyze trends by weekday.
- **Intensity**: Categorized activity levels (Low, Moderate, High) based on steps taken.

4. Data Models

The following tools and models were used:

- **NumPy and Pandas**: For data manipulation, cleaning, and statistical summaries.
- **Matplotlib**: For visualizing trends and relationships.

5. Approach

1. **Importing Libraries**: Used Pandas, NumPy, and Matplotlib for analysis and visualization.
2. **Data Reading and Cleaning**:
    - Converted ActivityDate to datetime format.
    - Checked for missing values and filled them appropriately.
    - Removed duplicate rows for cleaner analysis.
3. **Visualization**:
    - Histogram of daily step count to analyze the frequency distribution.
    - Line plot for calories burned over time to identify trends.
    - Scatter plot to analyze the correlation between steps and calories burned.
    - Bar charts for activity patterns by day of the week and intensity levels.
4. **Analysis and Insights**: Identified patterns in user activity and derived conclusions.

6. Project Results

- **Daily Activity**: The average number of steps and calories burned revealed typical user activity levels.
- **Weekly Trends**: Activity tends to peak on weekends, suggesting users are more active during leisure days.
- **Intensity Distribution**: Most users fall into the low to moderate intensity category, highlighting an opportunity for increased activity.
- **Correlation**: A strong positive correlation exists between steps and calories burned, confirming that increased movement leads to higher energy expenditure.

7. Conclusion

This analysis provided insights into user activity patterns and highlighted areas for improvement. Most users showed a need for increased activity, particularly during weekdays. The relationship between steps and calories emphasized the importance of staying active for better health outcomes. These findings can help users optimize their fitness routines and serve as valuable input for fitness tracker product enhancements.